

BRINGING TACTILE FEEDBACK TO VIRTUAL REALITY: THE WIRE AND HOOP GAME WITH BLUETOOTH-ENABLED HAPTICS



Authors & Advisor

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INTRODUCTION

Haptic feedback improves realism in VR but is often underused [1]. We designed a wearable haptic bracelet to provide 3-axis directional vibrations in a VR wire-and-hoop game. The goal was to support user navigation and reduce error rates.

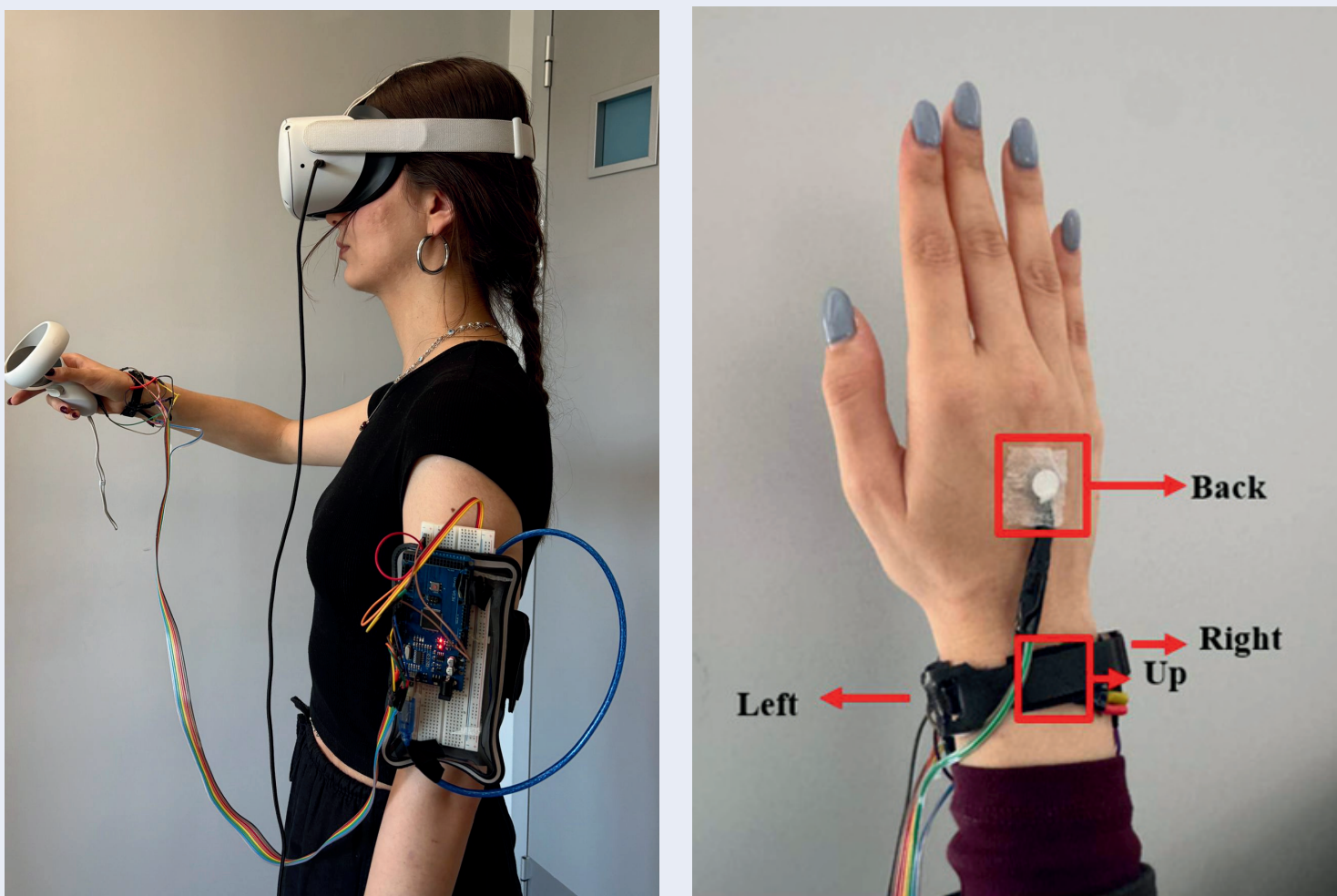


Figure 1. (A) The wearable haptic feedback bracelet and the armband worn by the user. (B) Motor positions seen from the upper surface of the hand.

HARDWARE DESIGN

- Six ERM vibration motors positioned on X, Y, Z axes
- Arduino Mega 2560 with HC-05 Bluetooth module.
- Directional feedback triggered via Unity distance detection.
- Powered by powerbank integrated into wearable armband for mobility.

RESULTS

Partial haptic feedback significantly reduced errors compared to full or no feedback. Handle size also impacted performance, the small handle caused more hits. Task duration was not significantly affected

CONCLUSION

Moderate, context-sensitive haptic feedback improves VR task accuracy without adding cognitive load [2]. Users preferred partial haptic feedback for being helpful yet non-intrusive. Ergonomic design and feedback levels are key factors in user performance.

Affiliations

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OBJECTIVE

To develop and evaluate a wearable haptic system that improves user performance in VR tasks through directional tactile feedback.

METHODOLOGY

- Participants: 18 students (9F/9M), aged 20–25
- Materials: Oculus Quest 2, Unity software, Bluetooth-enabled custom designed haptic bracelet.
- Procedure: Users completed tasks under 3 haptic conditions (none, partial, full) and 2 handle sizes (big, small)
- Data Analysis: ANOVA on error count & time; user feedback rated with 7-point Likert scale

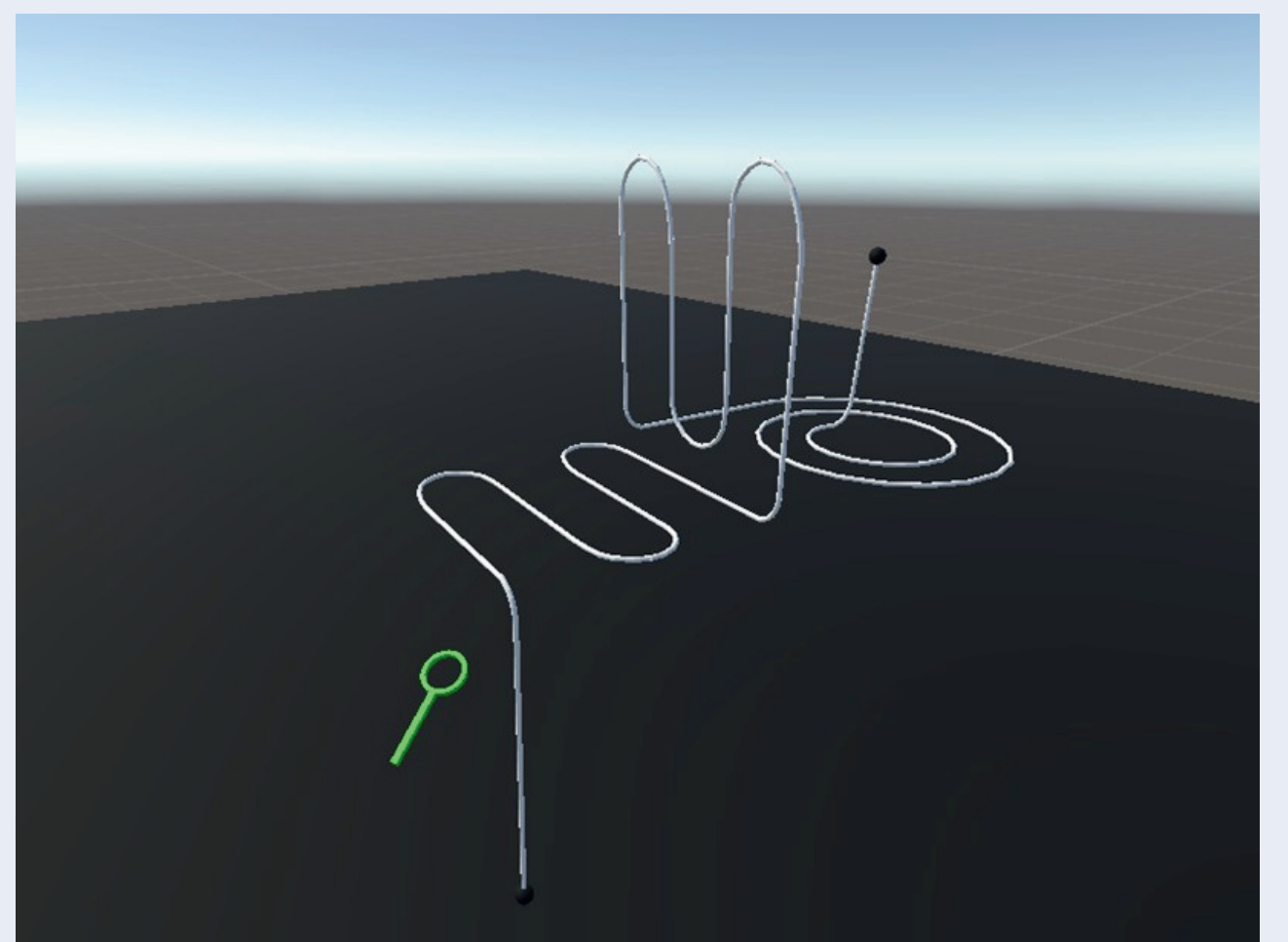


Figure 2. Virtual wire and handle setup in Unity.

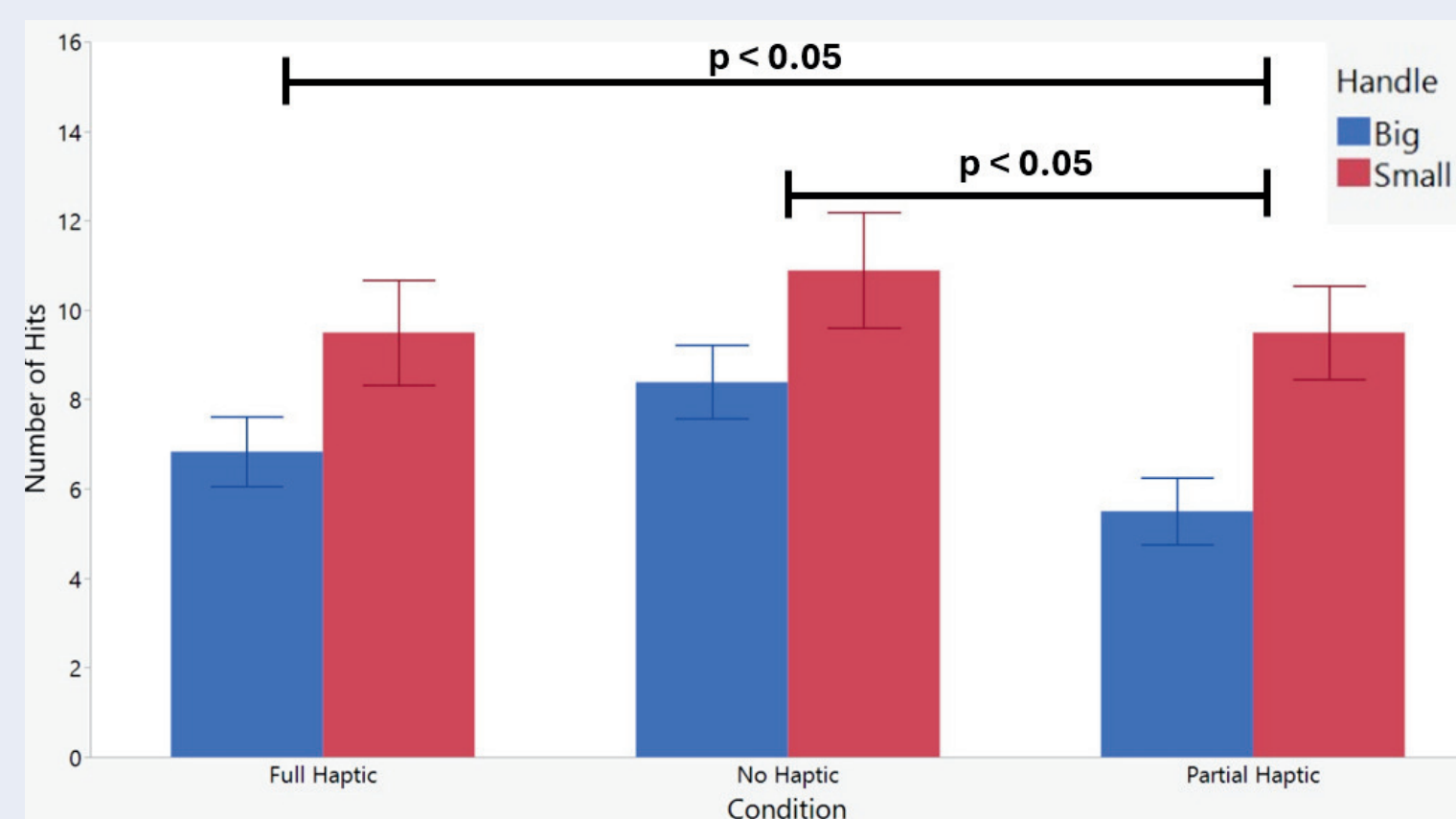


Figure 3. Number of Hits & Condition graph.

REFERENCES

- [1] Slater, M., & Sanchez-Vives, M. V. (2016). Enhancing our lives with immersive virtual reality. *Frontiers in Robotics and AI*, 3, 74.
- [2] Sigrist, R., Rauter, G., Riener, R., & Wolf, P. (2013). Augmented visual, auditory, haptic, and multimodal feedback in motor learning: A review. *Psychonomic Bulletin & Review*, 20(1), 21–53.